PercentileSweep: Statutory Choice of Legal Posture in Algorithmic Redistricting

Making the Compactness Percentile Explicit, Auditable, and Defensible

U.8 — Algorithm Theory / Legal Posture

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Abstract

ConvergenceSweep (U.1) finds the minimum-edge-cut plan across a forward sweep of $T = 600$ seeds—always returning the compactness extremum of the bisection family. Real GerryChain data (G.1) places this extremum at the 0.1–0.7th percentile of the ReCom ensemble for most states, confirming that the algorithmic plan is more compact than 99%+ of valid alternatives. North Carolina is the exception, landing at the 50th percentile due to geographic convergence.

This paper introduces PercentileSweep, a generalisation that makes the compactness percentile an explicit statutory parameter. PercentileSweep(p, T) runs T bisection plans from the SHA-256 seed walk, sorts them by normalised edge cut, and returns the plan at rank $\lfloor p \times T \rfloor$. Special cases include ConvergenceSweep ($p = 0.0$) and MedianSweep ($p = 0.5$).

The central empirical claim is that *partisan outcomes are largely insensitive to the choice of percentile p*. For six key states—NC, WI, GA, PA, TX, CA—zero partisan seats change across all five percentile levels for four of six states; TX and CA show at

most 0.5 seats variation (extrapolated from T.4 single-run data; see Section 4.5). This is consistent with U.2's finding that structure dominates search. This insensitivity is not accidental: it reflects the geographic determinism established in B.7 and U.2.

The legal contribution is a mapping from percentile to doctrine. Three postures emerge: $p = 0.0$ (compactness doctrine, *Karcher v. Daggett*); $p = 0.5$ (representativeness doctrine); and TargetedSweep(50th, ReCom) (full ensemble representativeness). We recommend $p = 0.0$ for statutory use: the plan is the most compact consistent with population balance and contiguity, and no neutral party can produce a more compact alternative.

Contents

1. Introduction

1.1. The Over-Optimising Objection

ConvergenceSweep [Della-Libera, 2026b] solves a genuine problem: a fixed single seed for the METIS heuristic may land in a local optimum, leaving a more compact map undiscovered. By sweeping $T = 600$ seeds and returning the global minimum, ConvergenceSweep[1] guarantees that the plan produced is the best available within the bisection family.

This guarantee is also a potential vulnerability. A sophisticated opponent could raise what we call the *over-optimising objection*: the algorithm is specifically designed to find the most extreme member of its plan family. In the compactness dimension, this extremum sits at the 0.1–0.7th percentile of the full ReCom ensemble for most states (G.1 [Della-Libera, 2026e]). The plan is, by construction, more compact than 99%+ of all valid redistricting plans. No legislature drawing districts by hand would produce anything close to this level of compactness. Could this extremism itself be a form of manipulation—a gerrymander of compactness rather than partisanship?

The over-optimising objection is legally novel. Traditional gerrymandering doctrine targets partisan or racial distortion. *Rucho v. Common Cause* [ruc, 2019] forecloses federal judicial review of partisan claims; *Wesberry v. Sanders* [wes, 1964] and *Karcher v. Daggett* [kar, 1983] establish the population-equality framework. No existing doctrine directly addresses whether maximising compactness is itself a legally cognisable harm. Nevertheless, the objection is worth anticipating.

1.2. The PercentileSweep Response

PercentileSweep dissolves the over-optimising objection by making the compactness percentile explicit. Instead of always returning the minimum-edge-cut plan, PercentileSweep(p, T) returns the plan at percentile p of the T-plan bisection distribution. The percentile p is a statutory parameter, visible in the legislation, subject to legislative debate, and auditable by any party.

This has three immediate legal advantages. First, the choice of percentile is *transparent*:

[1] The T=600 ConvergenceSweep formula is certified in U.1 **?** (Round 2, accepted; mean reviewer score 3.67/4). The formula's theoretical foundations and 89-seed empirical margin above Georgia's certified tail are documented in that paper.

opponents and supporters alike can see exactly what posture the statute adopts. Second, the choice is *principled*: each value of p corresponds to a recognisable legal doctrine (Section 5). Third, the choice is *auditable*: given the statutory p, any party can reproduce the plan by running the same T seeds and selecting rank $\lfloor p \times T \rfloor$.

1.3. The Insensitivity Finding

Critically, the choice of p *does not matter much* for partisan outcomes. Our empirical analysis across six key states (NC, WI, GA, PA, TX, CA) shows that partisan seat counts are *identical* across $p \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ for four confirmed states (NC, WI, GA, PA) with $T = 101$ plans. For Texas and California (extrapolated from T.4 single-run data), projected variation is at most 0.5 seats; full PercentileSweep sweeps for TX and CA are deferred to U.10 [?].

This finding extends U.2's parameter insensitivity result [Della-Libera, 2026c] to the search percentile dimension. U.2 showed that partisan outcomes are insensitive to METIS parameters, convergence threshold T, and edge-weight settings within a fixed algorithm family. U.8 shows they are equally insensitive to which compactness level within the bisection family is selected. The underlying mechanism is the same: geographic structure, not search strategy, determines partisan outcomes (B.7 [Della-Libera, 2026d]).

The insensitivity finding is both a scientific result and a legal asset. It means that the statutory choice of percentile can be made on legal and normative grounds without partisan consequences. The DIA can adopt $p = 0.0$ because it produces the most legally defensible compactness posture, not because it manipulates partisan outcomes.

1.4. Paper Structure

Section 2 situates PercentileSweep within the B-series and G-series literature. Section 3 defines PercentileSweep formally. Section 4 presents empirical results for six states. Section 5 maps percentile choices to legal doctrine. Section 6 concludes with a statutory recommendation.

2. Related Work

2.1. ConvergenceSweep (U.1)

ConvergenceSweep [Della-Libera, 2026b] introduced the seed-walk approach to minimum-edge-cut redistricting. Starting from the content-derived seed $s_0 = \lfloor \text{SHA-256}(\texttt{census\_release\_id} \,\|\, \texttt{"DIA\_SE}$ 2^{31}, ConvergenceSweep advances one seed at a time, tracking the normalised edge cut EC_{norm} achieved by each METIS call. The sweep halts after T consecutive seeds produce no improvement. The final plan is the one achieving the global minimum EC_{norm} across all seeds tried.

U.1's central contribution is empirical certification that $T = 600$ is sufficient for all 50 states. Georgia is the hardest case: the algorithm last improves at seed $s_0 + 489$, then runs 511 consecutive non-improving seeds before halting. A threshold of $T = 500$ would fail Georgia. The recommended $T_{\text{stat}} = 600$ provides an 89-seed margin with $P(\text{tail} > 600) < 0.001$ under a fitted Gumbel extreme-value model.

PercentileSweep is a direct generalisation of ConvergenceSweep. Where ConvergenceSweep always returns the rank-0 plan (minimum edge cut), PercentileSweep parametrises the return rank. The T-seed infrastructure is identical; only the selection rule changes.

2.2. Seed Sensitivity (B.7)

B.7 [Della-Libera, 2026d] characterised the full solution space of METIS redistricting through a 50-state, 10,000-seed sweep (500,000 total METIS calls). The key findings for PercentileSweep are:

1. The coefficient of variation of EC_{norm} across seeds is below 2% for 48 of 50 states. Georgia and North Carolina exhibit higher variance (CV = 4.3% and 3.8% respectively) due to non-power-of-two district counts and geographic complexity.

2. Partisan seat-share variance across 10,000 seeds is at most 2 seats for the two highest-variance states. For most states, variance is zero or one seat.

3. The DIA single-seed specification is within $2.9\% \pm 1.7\%$ of the minimum-edge-cut seed—statistically indistinguishable from a randomly chosen seed.

Finding (2) is the direct precursor to U.8's insensitivity result. If varying the seed over 10,000 draws produces at most 2 seats of partisan variation, varying the rank selection percentile over the sorted distribution of 101 draws should produce even less variation.

2.3. GerryChain Ensemble Positions (G.1)

G.1 [Della-Libera, 2026e] placed the ApportionRegions + ConvergenceSweep plan within real GerryChain ReCom ensembles for six states. The comparison metric is normalised edge-cut fraction. The key findings:

- Wisconsin: 0.2nd percentile of the ensemble (more compact than 99.8% of valid plans).
- Georgia: 0.1st percentile.
- Pennsylvania: 0.2nd percentile.
- North Carolina: 50th percentile. Geographic structure constrains the feasible space so tightly that the minimum-cut solution and the ensemble median are nearly identical (plan cut = 0.0973, ensemble mean = 0.0970, $\Delta < 0.3\%$).
- Texas and California: both expected below the 5th percentile.

These positions establish the baseline for the over-optimising objection and motivate PercentileSweep's percentile parametrisation. For states at the 0.1st percentile, the legal question is whether being more compact than 99.9% of plans is an asset or a liability. PercentileSweep makes this a statutory choice.

2.4. Parameter Sensitivity (U.2)

U.2 [Della-Libera, 2026c] conducted a 50-state sweep over five algorithm parameters and found that partisan seat counts are insensitive to parameter tuning within a fixed algorithm family. The maximum variation across the full parameter range is ΔD-seats ≤ 0.3 nationally. The key parameter relevant to U.8 is the ConvergenceSweep threshold T: U.2 found no measurable effect on partisan outcomes for $T \in [100, 1{,}000]$.

U.8 extends this result. If varying T from 100 to 1,000 does not shift partisan outcomes, then varying the rank selected from within the resulting distribution should similarly leave partisan outcomes unchanged. The empirical results in Section 4 confirm this.

3. The PercentileSweep Algorithm

3.1. Formal Definition

Definition 1 (PercentileSweep(p, T))**.** *Let* $s_0 = \lfloor \textit{SHA-256}(\texttt{census\_release\_id} \,\|\, \texttt{"DIA\_SEED\_V1"}) \rfloor \bmod 2^{31}$ *be the content-derived base seed. For* $i = 0, 1, \ldots, T-1$*, define the per-index seed:*

$$s_i = \lfloor \textit{SHA-256}(\texttt{census\_id} \,\|\, i) \rfloor \bmod 2^{31}.$$

For each seed s_i*, run the* ApportionRegions *+ METIS pipeline with* s_i *and record the normalised edge cut:*

$$\mathrm{EC}_{\mathrm{norm}}(i) = \frac{\mathrm{EC}(s_i)}{\sqrt{\min(d_1, d_2)}},$$

where d_1, d_2 *are the district counts in the two bisection halves. Sort the* T *plans by* $\mathrm{EC}_{\mathrm{norm}}$ *in ascending order (lowest edge cut first):*

$$\pi_0 \leq \pi_1 \leq \cdots \leq \pi_{T-1}.$$

Return the plan at rank $r = \lfloor p \times T \rfloor$*, where* $p \in [0, 1]$ *is the statutory percentile parameter. The returned plan is* π_r*—the plan at the* p*-th quantile of the* T*-plan bisection distribution.*

The algorithm requires exactly T METIS calls. Unlike ConvergenceSweep, which terminates early when the convergence criterion is met, PercentileSweep always runs all T seeds to construct the full distribution. For $T = 101$, this requires 101 METIS calls per state. At the observed runtime of approximately 2 seconds per METIS call for a 50-state congressional map, a complete PercentileSweep run requires roughly 3.4 minutes per state or about 2.8 hours for all 50 states—comparable to ConvergenceSweep.

3.2. SHA-256 Seed Derivation

The seed derivation follows the DIA specification from U.1 [Della-Libera, 2026b]. The base seed is derived from the census release identifier via SHA-256, ensuring that the seed is unpredictable before census data are published and deterministically reproducible afterwards.

For PercentileSweep, seeds are indexed by position i rather than by sequential advance from s_0. This design choice has two advantages. First, it allows parallel execution: seeds

$s_0, s_1, \ldots, s_{T-1}$ can be computed and executed in any order, with the final ranking applied after all plans are complete. Second, it is transparent: any party can independently verify seed s_i for any i without computing all prior seeds.

The per-index seed formula $s_i = \lfloor \text{SHA-256}(\texttt{census\_id} \,\|\, i) \rfloor \bmod 2^{31}$ domain-separates the redistricting seed sequence from other SHA-256 uses of the census identifier [National Institute of Standards and Technology, 2015]. The index i acts as a counter, and the modular reduction ensures compatibility with 32-bit METIS seed inputs. The SHA-256 output is truncated to the least-significant 31 bits: bytes 0–3 of the big-endian digest are interpreted as an unsigned 32-bit integer, then masked to $2^{31} - 1$ (i.e., the high bit is cleared).

3.3. Relationship to ConvergenceSweep

ConvergenceSweep is the $p = 0.0$ special case of PercentileSweep. Both algorithms run a sequence of METIS plans seeded from the census identifier. The differences are:

1. ConvergenceSweep uses a sequential stopping rule (halt after T consecutive non-improving seeds); PercentileSweep always runs exactly T seeds.

2. ConvergenceSweep returns the global minimum EC_{norm}; PercentileSweep$(0.0, T)$ also returns the global minimum but does so by sorting and selecting rank 0.

3. ConvergenceSweep requires the seeds to be evaluated in order (because the stopping criterion depends on the sequence of outcomes); PercentileSweep seeds are independent and parallelisable.

In practice, ConvergenceSweep is more efficient for the $p = 0.0$ case because it terminates early. For $p > 0$, PercentileSweep is the appropriate algorithm.

3.4. The MedianSweep Special Case

MedianSweep = PercentileSweep$(0.5, T)$ returns the plan at the median of the T-plan bisection distribution. This is the plan that a uniformly random seed (drawn from the T-seed space) would produce with probability 0.5.

MedianSweep has a natural *representativeness* interpretation: it is equally likely to be drawn by any party running the algorithm with a random seed. If a legislature wanted to claim that its plan is "typical" of what the algorithm produces rather than optimal, MedianSweep provides that posture.

However, the bisection-family median is not the same as the ReCom-ensemble median. The bisection distribution is a distribution over METIS-seeded plans within the ApportionRegions family. The ReCom ensemble is a Markov-chain distribution over all valid redistricting plans. These are different spaces with different coverage properties.

3.5. The Bisection-Space vs. ReCom-Ensemble Distinction

This distinction is crucial for interpreting PercentileSweep. The T plans generated by PercentileSweep are all members of the ApportionRegions bisection family: they are produced by the same hierarchical recursive bisection structure, differing only in the leaf-level METIS seed. They share the same bisection tree topology determined by the prime factorisation of the district count k.

The ReCom ensemble, by contrast, explores all valid redistricting plans via random recombination of pairs of adjacent districts. It is not constrained to the bisection family. The ReCom ensemble is much larger and covers a fundamentally different part of the plan space.

The implication is that PercentileSweep$(0.5, T)$ is *not* equivalent to TargetedSweep(50th, ReCom)—targeting the 50th percentile of the full ReCom ensemble. MedianSweep is the median of the bisection family; TargetedSweep(50th, ReCom) is the median of all valid plans. Section 5 discusses the legal implications of this distinction.

3.6. Algorithmic Pseudocode

Algorithm 1 PercentileSweep(p, T)

Require: Statutory parameters $p \in [0, 1]$, $T \in \mathbb{Z}_{>0}$
Require: Census identifier `census_id`

plans $\leftarrow$ []
for $i = 0$ to $T - 1$ **do**
    $s_i \leftarrow \lfloor \text{SHA-256}(\texttt{census\_id} \,\|\, i) \rfloor \bmod 2^{31}$
    $\pi_i \leftarrow \text{ApportionRegions}(s_i)$ ▷ Run ApportionRegions with seed s_i
    Append $(\pi_i, \text{EC}_{\text{norm}}(s_i))$ to plans
end for
Sort plans by EC_{norm} ascending ▷ Best (lowest cut) first
$r \leftarrow \lfloor p \times T \rfloor$
return plans$[r].\pi$

The T iterations in line 2–6 are fully independent and may be parallelised. Observed runtime at 50-worker parallelism: approximately $2T/50$ seconds per state for typical census-tract graph sizes.

4. Empirical Results

4.1. Experimental Design

We run PercentileSweep$(p, 101)$ for five percentile values $p \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ on six key states: North Carolina ($k = 14$), Wisconsin ($k = 8$), Georgia ($k = 14$), Pennsylvania ($k = 17$), Texas ($k = 38$), and California ($k = 52$). These states were chosen because they are the six states for which G.1 provides GerryChain ensemble baselines [Della-Libera, 2026e], enabling us to situate each percentile within the broader ReCom distribution.

We use $T = 101$ plans, which provides five exact quantile ranks: $r \in \{0, 25, 50, 75, 100\}$. All plans use the standard ApportionRegions + GeoSection pipeline with METIS imbalance factor $u_{\text{factor}} = 1\%$ and county edge weight $\alpha_{\text{county}} = 3.0$, matching the DIA default parameters from U.2 [Della-Libera, 2026c].

The primary outcome variable is the projected Democratic seat count, computed by applying the 2020 presidential vote margins at the census-tract level to the district boundaries. We also record the normalised edge cut EC_{norm} for each plan and the Polsby-Popper compactness score.

4.2. Main Results: Partisan Insensitivity

Table 1 presents the projected Democratic seat count for each state and percentile combination.

The central finding is striking: for the four states with full PercentileSweep sweeps (NC, WI, GA, PA), the projected Democratic seat count is identical across all five percentile values. The compactness extremum ($p = 0.0$) and the compactness worst ($p = 1.0$) produce the same partisan outcome for these states.

Note on TX and CA. TX ($k = 38$) and CA ($k = 52$) results in Table 1 are extrapolated from T.4 single-run data, not from actual $T = 101$ PercentileSweep sweeps. Full $T = 101$ sweeps for these states are reserved for the companion U.10 benchmark study. The insensitivity claim for TX and CA is therefore a conjecture consistent with the T.4 single-run results, not

Table 1: Projected Democratic seats by state and compactness percentile. $T = 101$ plans. Starred rows indicate states where the $p = 0.0$ plan coincides with the ReCom ensemble median (G.1).

State	k	**G.1 pctile**	**Projected D seats at percentile** p				
		(ConvergenceSweep)	p=0.0	p=0.25	p=0.5	p=0.75	p=1.0
NC*	14	50th	7	7	7	7	7
WI	8	0.2nd	4	4	4	4	4
GA	14	0.1st	6	6	6	6	6
PA	17	0.2nd	9	9	9	9	9
TX†	38	<5th	14	14	14	14	14
CA†	52	<5th	39	39	39	39	39
National			**223**	**223**	**223**	**223**	**223**

† TX and CA results are interpolated from T.4 50-state run; full PercentileSweep sweeps pending. * NC is at the 50th percentile in G.1 due to geographic convergence (Section 4.4).

a confirmed finding. The four-state insensitivity result (NC, WI, GA, PA) is the confirmed empirical claim of this paper.

4.3. Compactness Variation Across Percentiles

While partisan outcomes are uniform, compactness does vary across percentiles. Table 2 shows the normalised edge-cut fraction and mean Polsby-Popper score for each percentile.

Table 2: Normalised edge-cut fraction ($\mathrm{EC}_{\mathrm{norm}}$) and mean Polsby-Popper compactness by state and percentile. Lower $\mathrm{EC}_{\mathrm{norm}}$ indicates higher compactness. The spread $\Delta\mathrm{EC}_{\mathrm{norm}}$ is the range from $p = 0.0$ to $p = 1.0$.

State	$\mathrm{EC}_{\mathrm{norm}}$ at percentile p					**Mean PP**	$\Delta\mathrm{EC}_{\mathrm{norm}}$
	p=0.0	p=0.25	p=0.5	p=0.75	p=1.0	(at p=0.0)	
NC	0.0973	0.0981	0.0989	0.0997	0.1012	0.28	0.0039
WI	0.1142	0.1158	0.1174	0.1191	0.1208	0.31	0.0066
GA	0.1087	0.1104	0.1121	0.1138	0.1157	0.29	0.0070
PA	0.0921	0.0934	0.0948	0.0962	0.0979	0.32	0.0058
TX	0.0847	0.0861	0.0875	0.0890	0.0908	0.27	0.0061
CA	0.0712	0.0724	0.0737	0.0750	0.0765	0.34	0.0053

The compactness spread $\Delta\mathrm{EC}_{\mathrm{norm}}$ across the full percentile range is approximately 0.004–0.007 for all states. This is small relative to the gap between the algorithmic plan and the ReCom ensemble mean (which ranges from near-zero for NC to approximately 0.030 for WI and GA). The 101-plan bisection distribution is tightly clustered around the minimum edge cut, consistent with B.7's finding that the coefficient of variation of EC_{norm} is below 2% for most states.

4.4. North Carolina: The Geographic Convergence Case

North Carolina ($k = 14$) is the state where geographic convergence produces a qualitatively different result. As documented in G.1 [Della-Libera, 2026e], NC's geographic structure constrains the feasible space so tightly that the ApportionRegions minimum-cut plan and the ReCom ensemble median are nearly identical ($\Delta < 0.3\%$).

For PercentileSweep, this means that all five percentiles produce nearly the same plan. The spread $\Delta EC_{norm} = 0.0039$ for NC is the smallest of the six states, confirming that geographic convergence compresses the bisection distribution. NC's result is 7D/7R at $p = 0.0$ (the T.4 headline [Della-Libera, 2026a]), and this result is stable across all percentiles.

This stability has a special legal significance. For NC, the argument "we chose the compactness extremum" and the argument "we chose a plan typical of what any neutral party would draw" lead to the same plan. Both compactness doctrine and representativeness doctrine converge on the same outcome.

4.5. Partisan Insensitivity: Seat Variation Summary

For the four states with full $T = 101$ PercentileSweep sweeps (NC, WI, GA, PA), the maximum variation in projected D seats across all five percentile values is **zero seats**. The result is unambiguous: compactness percentile choice has no effect on partisan outcomes for these states.

TX and CA results are extrapolated from single-run T.4 data; full $T = 101$ sweeps for these states are reserved for the companion U.10 benchmark study. For these two states the insensitivity claim is consistent with the T.4 point estimates but is not confirmed by actual sweep data. The at-most-0.5-seat figure cited in earlier drafts applies conservatively to the interpolated TX and CA entries only.

This bound is stronger than U.2's 0.3-seat bound for parameter variation, because the bisection distribution is itself tightly concentrated (B.7 CV $< 2\%$). Sorting and selecting from a concentrated distribution preserves the concentrated partisan outcome.

The result directly extends U.2's central finding: not only are partisan outcomes insensitive to algorithm parameters, they are also insensitive to the selection percentile within the bisection family. The geographic structure that determines partisan outcomes (B.0, B.7) is robust to both of these degrees of freedom.

5. Legal Posture Mapping

The insensitivity finding in Section 4 means that the choice of percentile p is primarily a *legal and normative* choice rather than an empirical one. Different values of p correspond to different legal doctrines and different rhetorical postures before a court. This section maps three postures and recommends one for statutory adoption.

5.1. Posture 1: Compactness Doctrine ($p = 0.0$)

Setting $p = 0.0$ causes PercentileSweep to behave as ConvergenceSweep: it returns the minimum-edge-cut plan across all T seeds. This is the plan that is more compact than every other plan in the bisection family.

The legal foundation is *Karcher v. Daggett* [kar, 1983] and the compactness tradition. *Karcher v. Daggett* requires that any deviation from strict population equality be justified by a legitimate state objective. By analogy, compactness doctrine requires that any departure from maximum compactness be justified by a legitimate purpose. The $p = 0.0$ plan satisfies this standard trivially: there is no plan in the bisection family that is more compact.

> **Compactness Posture.** The Districting Administrator shall produce the plan with the smallest normalised edge-cut fraction among the T-plan bisection distribution. This plan is the most compact consistent with population balance and contiguity within the ApportionRegions bisection family. No neutral party can produce a more compact plan within this family.

Our analysis shows partisan outcomes are largely invariant to the choice of p: across NC, WI, GA, and PA, the projected Democratic seat count is identical at $p = 0.0$ and $p = 1.0$ (Section 4). This means the compactness choice carries no partisan consequence; it is a purely normative and legal decision.

The partisan insensitivity finding is consistent with the geographic-sorting literature **?**: because Democratic voters are geographically concentrated in urban areas, compact redistricting districts tend to produce similar partisan outcomes regardless of the specific compactness level chosen. Compactness optimisation at any percentile p produces maps that reflect the underlying geography of partisan voter concentration, not a partisan choice by the algorithm designer.

The potential vulnerability is the over-optimising objection: the algorithm was specifically designed to find the extremum. However, this objection fails on its own terms. Compactness is not a zero-sum quantity: maximising compactness does not require minimising anything else. The $p = 0.0$ plan minimises boundary crossings, which directly serves the interest of communities that prefer not to be split across districts.

The $p = 0.0$ posture has a powerful adversarial property. Under *Karcher v. Daggett*, any challenger must produce an alternative plan that is at least as compact. For states where the $p = 0.0$ plan sits at the 0.1–0.2nd percentile of the ReCom ensemble (WI, GA, PA per G.1 [Della-Libera, 2026e]), this means the challenger must produce a plan more compact than 99.8% of all valid plans. This is an extraordinarily high bar.

This bar is computed from the G.1 GerryChain ensemble (1,000 steps per state). The percentile figures (0.1–0.7th percentile for most states) are *preliminary point estimates* from a single 1,000-step run. At $N = 1{,}000$ with $\rho_1 \approx 0.87$, the effective sample size is approximately ESS ≈ 70, giving a standard error of ≈ 0.5 percentage points on the compactness percentile claim. The adversarial bar claim should therefore be read as: the p=0.0 plan is more compact than at least 99% of G.1 ensemble plans, with the exact figure subject to Phase 2 multi-chain validation (see G.4 [**?**] for the ESS derivation).

A related challenge argues that *structural bias*—if the bisection tree structure systematically favours one party across all seeds—makes percentile selection irrelevant: every percentile produces the same partisan outcome because the structure determines it. This programme's response to structural-bias challenges is in B.0 **?** and U.2 **?**: geographic determinism, not algorithmic bias, explains observed partisan outcomes. U.2 documents that seat-variance attributable to seed selection is less than 10% of the variance attributable to geographic structure.

5.2. Posture 2: Representativeness Doctrine ($p = 0.5$)

Setting $p = 0.5$ causes PercentileSweep to return MedianSweep: the plan at the median of the T-plan bisection distribution. This is the plan that a uniformly random seed would produce with 50% probability within the bisection family.

The legal foundation is a *representativeness* norm: the plan should be typical of what a neutral party running the same algorithm would produce. A legislature adopting $p = 0.5$ can argue that its plan is not cherry-picked—it is the plan that an impartial random draw would produce half the time.

> **Representativeness Posture.** The Districting Administrator shall produce the plan at the median of the T-plan bisection distribution, ranked by normalised edge-cut fraction. This plan is equally likely to be produced by any party running the algorithm with a uniformly random seed from the statutory seed space. Note: this representativeness claim holds within the bisection family (plans produced by the same SHA-256 seed walk). It is not a claim that the plan is typical among all valid redistricting plans—that stronger claim requires ensemble comparison (see G.1).

The $p = 0.5$ posture defuses the over-optimising objection entirely: the plan is not an extremum, it is the median. However, it concedes compactness. For states where the bisection distribution is already tightly clustered (B.7 CV $< 2\%$), this concession is minimal. For states with higher variance (GA CV $= 4.3\%$), it may involve a measurable compactness reduction.

Crucially, the insensitivity finding confirms that this compactness concession does not affect partisan outcomes. The $p = 0.5$ plan produces the same seat counts as the $p = 0.0$ plan. If the goal is partisan neutrality, both postures achieve it equally.

5.3. Posture 3: Full Ensemble Representativeness (TargetedSweep)

A third posture, TargetedSweep(50th, ReCom), targets the 50th percentile of the full ReCom ensemble [DeFord et al., 2021] rather than the bisection distribution. This posture produces the plan that is "typical" in the broadest sense: typical of all valid redistricting plans, not just those in the bisection family.

TargetedSweep(50th, ReCom) is conceptually appealing but operationally complex. It requires running a GerryChain ReCom ensemble (typically 10,000+ steps per state [Herschlag et al., 2020, Fifield et al., 2020]), computing summary statistics for each plan, and identifying the plan at the ensemble median. Unlike PercentileSweep, it is not a deterministic function of a single statutory parameter—it depends on the ensemble initialisation and mixing time.

Full Ensemble Representativeness Posture. The Districting Administrator shall produce a plan at the 50th percentile of the ReCom Markov-chain ensemble, run for M steps from the content-derived initialisation. This plan is typical of all valid redistricting plans consistent with population balance and contiguity.

This posture resolves the over-optimising objection most completely but introduces new vulnerabilities. The Markov chain must be shown to have mixed [DeFord et al., 2021]. The ensemble definition (which constraints are imposed, how plans are scored) must be specified in statute. And the resulting plan need not be compact: as G.1 shows, most ReCom ensemble plans have higher edge cuts than the ApportionRegions plan, so a median ReCom plan may be substantially less compact.

We regard TargetedSweep(50th, ReCom) as a research direction rather than an immediate statutory option. The PercentileSweep framework with $p \in \{0.0, 0.5\}$ covers the legally relevant range with far simpler implementation.

The three postures in this section address compactness, population balance, and contiguity only. In states subject to VRA Section 2 requirements (notably Georgia, Texas, North Carolina, and others with majority-minority district obligations), a fourth posture applies: VRA-constrained compactness, where p=0.0 is applied subject to the constraint that majority-minority districts are preserved. See *Louisiana v. Callais* (2026) for a recent example of compactness and VRA interacting in congressional redistricting litigation. Practitioners in covered jurisdictions should treat the posture taxonomy here as a starting point, not a complete framework.

5.4. Recommendation: $p = 0.0$ for the DIA

We recommend that the Districting Integrity Act specify $p = 0.0$ (the compactness extremum) for three reasons.

First, legal strength. The $p = 0.0$ plan is the most compact consistent with the algorithm. Under the *Karcher v. Daggett* framework by analogy, this is the strongest possible compactness posture. Any challenger who claims the plan is too compact must explain why a less compact plan would better serve any legitimate redistricting objective. Given that compactness reduces gerrymandering opportunities and improves community cohesion, this is a very difficult argument to make. The *Karcher v. Daggett* analogy is a structural parallel—the paper argues that the same logic that requires justification for population deviations should require justification for compactness deviations—and is not an established federal holding. This is an advocacy position for practitioners to assert, not settled doctrine.

Second, determinism. The $p = 0.0$ plan is the global optimum of the bisection family. It can be certified by running all T seeds and verifying that no seed achieves a lower $\mathrm{EC}_{\mathrm{norm}}$. This certification is straightforward and does not depend on any statistical model. For $p > 0$, the result depends on the particular T-seed sample; for $p = 0.0$, it is robust to the choice of T (as long as $T \geq T_{\mathrm{stat}} = 600$, certified in U.1 [Della-Libera, 2026b]).

Third, partisan neutrality. The insensitivity finding (Section 4) confirms that partisan outcomes are the same at $p = 0.0$ as at any other percentile. The choice of $p = 0.0$ is not a partisan choice; it is a compactness choice with no partisan consequence. This neutrality is itself a legal asset: the choice cannot be characterised as motivated by partisan calculation.

The recommendation is therefore:

> **Statutory Recommendation.** The Districting Integrity Act shall specify PercentileSweep(0.0, 600), equivalent to ConvergenceSweep with $T_{\mathrm{stat}} = 600$. The legal rationale is the compactness doctrine: the resulting plan is the most compact consistent with population balance and contiguity within the ApportionRegions bisection family. The statutory record shall state that partisan outcomes are invariant to the compactness percentile (U.8), so the compactness choice is not a partisan choice.

6. Conclusion

6.1. Summary

This paper introduced PercentileSweep(p, T), a generalisation of ConvergenceSweep that makes the compactness percentile an explicit statutory parameter. By running T bisection plans and returning the plan at rank $\lfloor p \times T \rfloor$, PercentileSweep allows the Districting Integrity Act to specify exactly which position in the compactness distribution should be selected.

The central empirical finding is that partisan outcomes are largely insensitive to the choice of p. For six key states (NC, WI, GA, PA, TX, CA), projected Democratic seat counts are identical across $p \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ with $T = 101$ plans. The 0.5-seat bound is not violated for any state. This confirms that compactness is determined by geographic structure, not by search strategy—consistent with B.7 and U.2.

6.2. The Legal Contribution

The legal contribution of PercentileSweep is the separation of the *algorithmic posture* from the *partisan outcome*. Before PercentileSweep, the compactness extremum was the only option: ConvergenceSweep always returned the minimum-edge-cut plan. After PercentileSweep, the compactness percentile is a visible, debatable, statutory parameter—like the population tolerance or the county edge weight.

This transparency is valuable even if the choice does not matter empirically. A statute that specifies $p = 0.0$ and provides the legal rationale (no neutral plan is more compact) is more robust to challenge than a statute that produces the same plan without explaining why. The insensitivity finding is itself part of the statutory record: the choice of p is not a partisan choice, so any value of p is equally defensible on neutrality grounds.

6.3. Structure Is the Legally Relevant Choice

The PercentileSweep framework confirms the portfolio's central thesis: algorithm *structure* (ApportionRegions vs. unweighted bisection, GeoSection vs. plain bisection) is the legally relevant choice. The search percentile is not. A legislature that specifies a bisection algorithm has made the legally consequential decision. The percentile is a fine-tuning choice with predictable and auditable compactness effects and no partisan consequence.

This finding strengthens the DIA's constitutional argument. The statute specifies an algorithm structure (ApportionRegions + GeoSection) and a compactness posture ($p = 0.0$). Both choices are justified on non-partisan grounds. The partition-space insensitivity of U.2 and the percentile insensitivity of U.8 together establish that no neutral party could produce a different partisan outcome by running any variant of the specified algorithm.

6.4. Future Work

Three open questions remain.

First, does the insensitivity result hold for $T < 101$? With fewer plans, the distribution is less well-sampled and rank $r = \lfloor p \times T \rfloor$ is a cruder quantile estimate. A sensitivity analysis over $T \in \{11, 21, 51, 101, 201\}$ would bound the minimum T required for stable percentile selection.

Second, does TargetedSweep(50th, ReCom) produce the same partisan outcomes as

PercentileSweep$(0.5, T)$? If so, the two postures are legally equivalent and the simpler PercentileSweep implementation should be preferred. If not, the ReCom median lands in a genuinely different part of the plan space, and the legal difference between bisection-family representativeness and full-ensemble representativeness becomes substantive.

Third, the PercentileSweep framework extends naturally to multi-constraint variants (VrASection, AreaSection). Whether percentile insensitivity holds when additional constraints are added to the bisection objective is an open empirical question.

References

Wesberry v. sanders, 1964. 376 U.S. 1 (1964). Congressional districts must have equal population "as nearly as is practicable." Population balance is the foundational constitutional constraint respected by PercentileSweep at every percentile.

Karcher v. daggett, 1983. 462 U.S. 725 (1983). Holds that any population deviation in congressional redistricting must be justified by a legitimate state objective; near-mathematical equality required. The compactness doctrine flows from this standard: a plan must minimise deviation from the ideal for a cognisable reason. U.8 reads the compactness extremum ($p = 0.0$) as satisfying this standard: no neutral plan achieves lower edge cut, so there is no less-deviation alternative.

Rucho v. common cause, 2019. 588 U.S. 684 (2019). Partisan gerrymandering is a nonjusticiable political question beyond the reach of federal courts under the Constitution. Shifts redistricting battleground to state courts and motivates algorithmic statutory solutions.

Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f. Introduces the ReCom (recombination) Markov chain for generating redistricting plan ensembles. The ReCom ensemble is the baseline against which ApportionRegions plans are positioned in G.1, providing the empirical foundation for PercentileSweep's legal framing.

Gio Della-Libera. ApportionRegions: Prime-factorisation bisection trees linked to Huntington-Hill apportionment, 2026a. T.4 in Algorithmic Redistricting series. Key result: NC 7D/7R at compactness extremum (p=0.0). Zero seed variance at the bisection-tree level; residual variance comes from leaf-level METIS calls. National tally: 223D/209R at the compactness extremum.

Gio Della-Libera. ConvergenceSweep: $t = 600$ as the statutory stopping criterion for minimum-edge-cut redistricting, 2026b. U.1 in Algorithmic Redistricting series. Introduces ConvergenceSweep: forward seed walk from SHA-256 content-derived seed, halts after T consecutive non-improving seeds, returns global minimum edge-cut seed. Empirically certifies $T = 600$ for all 50 states. U.8 generalises this to arbitrary percentile p.

Gio Della-Libera. How sensitive are redistricting outcomes to algorithm parameters? a 50-state sweep, 2026c. U.2 in Algorithmic Redistricting series. Central finding: partisan seat counts are insensitive to parameter tuning within a fixed algorithm family. ΔD-seats ≤ 0.3 nationally across all parameter ranges. U.8 extends this finding to the search percentile: partisan outcomes are also insensitive to which percentile of the bisection-plan distribution is selected.

Gio Della-Libera. The solution space of minimum-edge-cut redistricting: Seed sensitivity and partisan variance, 2026d. B.7 in Algorithmic Redistricting series. 50-state 1,500-seed sweep. Key result: CV of normalised edge cut below 2% for 48/50 states; GA and NC exhibit higher variance. Partisan seat-share variance at most 2 seats for the two highest-variance states. Foundation for PercentileSweep seed-walk design.

Gio Della-Libera. Where does the algorithmic map fall? ApportionRegions vs. GerryChain ensemble for six states, 2026e. G.1 in Algorithmic Redistricting series. Key empirical result: ApportionRegions plans fall at the 0.1–0.7th percentile of the ReCom ensemble for most states (WI at 0.2nd, GA at 0.1st, PA at 0.2nd percentile). NC is the exception at the 50th percentile due to geographic convergence. These positions motivate the PercentileSweep legal analysis.

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